Heaven's Light is our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

Project

Course Code : CSE-3100

Course Name : Web Based Application Project

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Series : 22	

Project Title:

Learnify: A Web-Based Learning Management System for Students and Educators

Introduction:

Online education is growing rapidly. As a result, digital learning platforms have become essential for students and educators. This project presents a web-based Learning Management System (LMS) that offers a simple and secure platform for online learning. The system supports course enrollment, video based learning, testing, assessments, and educator management, along with features such as gamification (badges and points) and mobile accessibility to enhance learner engagement.

Objective of the Project:

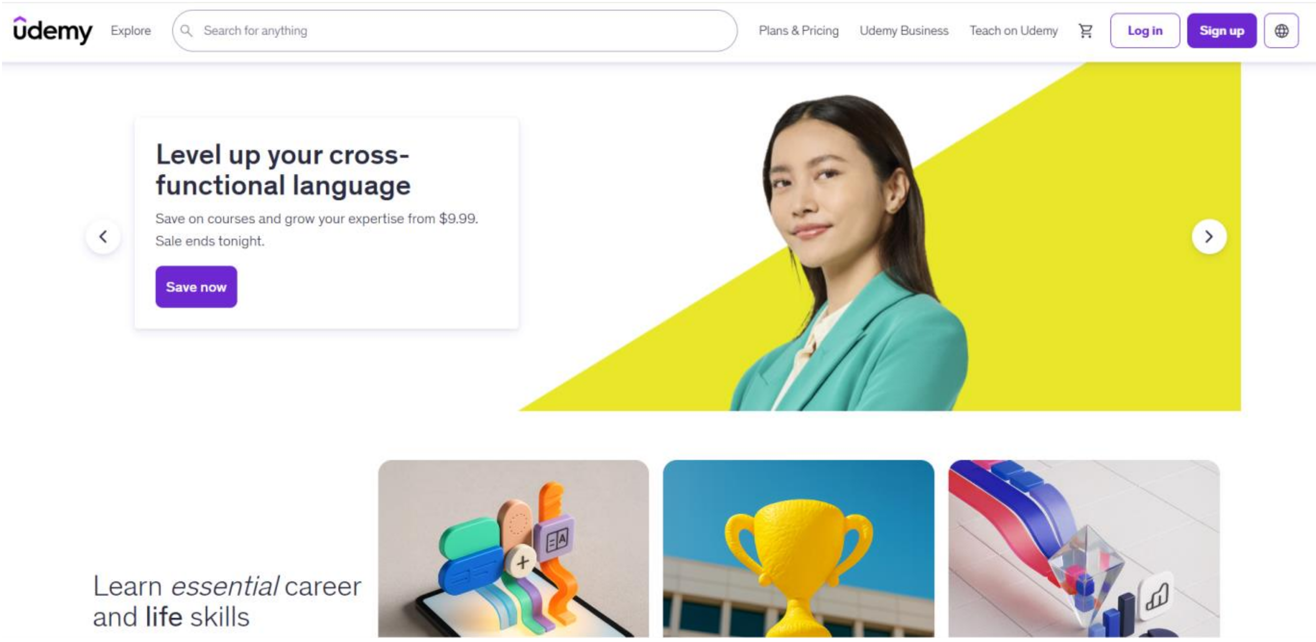
The main objectives of this project are:

1. To develop a web-based platform for online learning.
2. To allow students to enroll in and access online courses.
3. To provide educators with tools to create and manage courses.
4. To implement secure user authentication and payment processing.
5. Include Testing and Assessment feature such as quizzes.
6. To ensure mobile-friendly and responsive design.
7. To improve learning engagement compared to traditional learning management system platforms.

Literature Review:

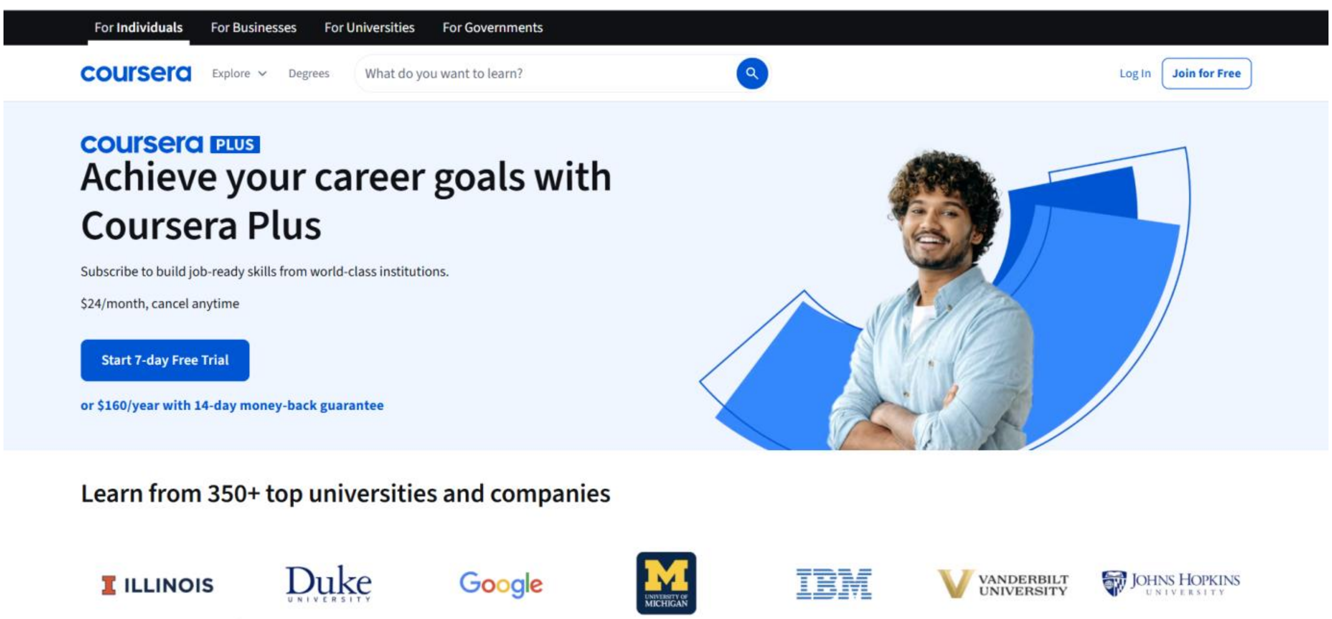
1. Udemy [1]

Introduction: An online course marketplace offering skill-based learning.
Features: Video courses, ratings, certificates, lifetime access.
Limitations: Limited structured learning paths and weak engagement features.



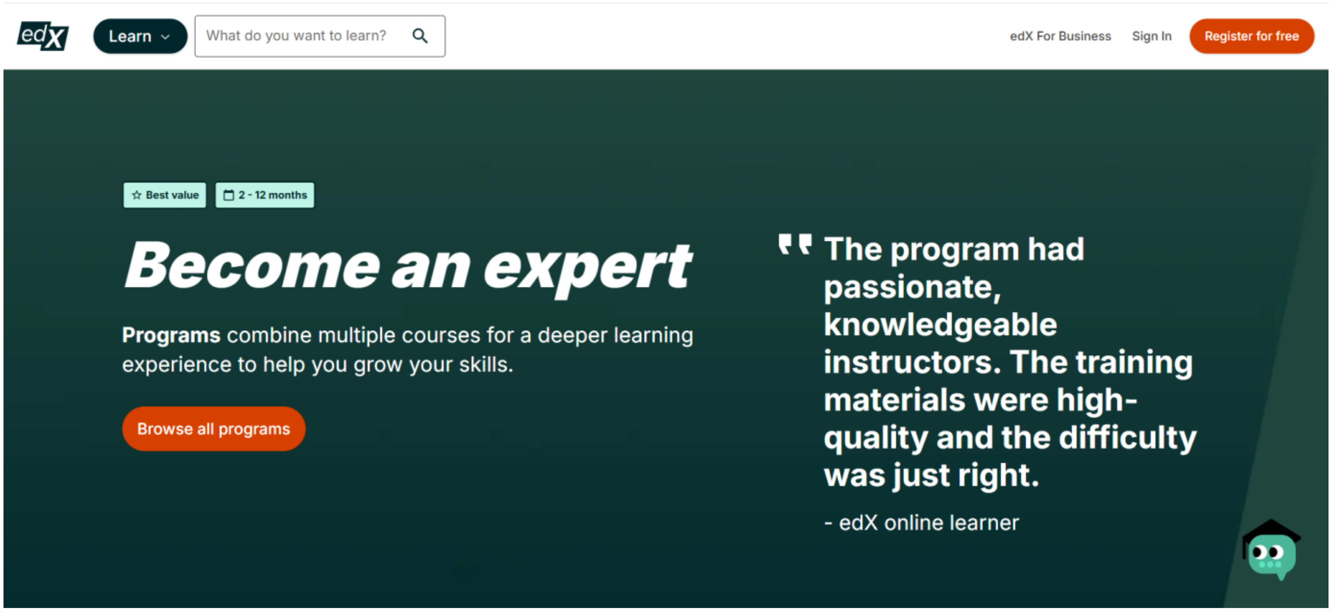
2. Coursera [2]

Introduction: Offers university-level online courses and certifications.
Features: Professional certificates, quizzes, graded assignments.
Limitations: High cost and complex interface for beginners.



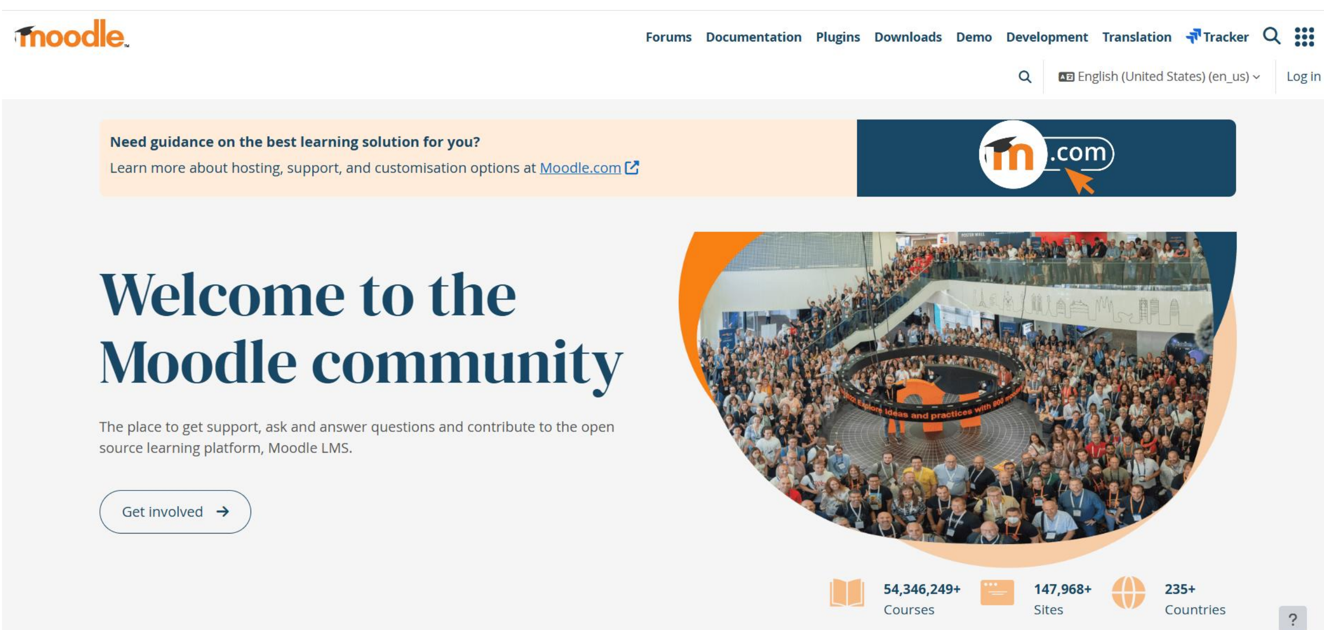
3. edX [3]

Introduction: Online education platform by universities and institutions.
Features: Certificates, degree programs.
Limitations: Limited personalization and engagement features.



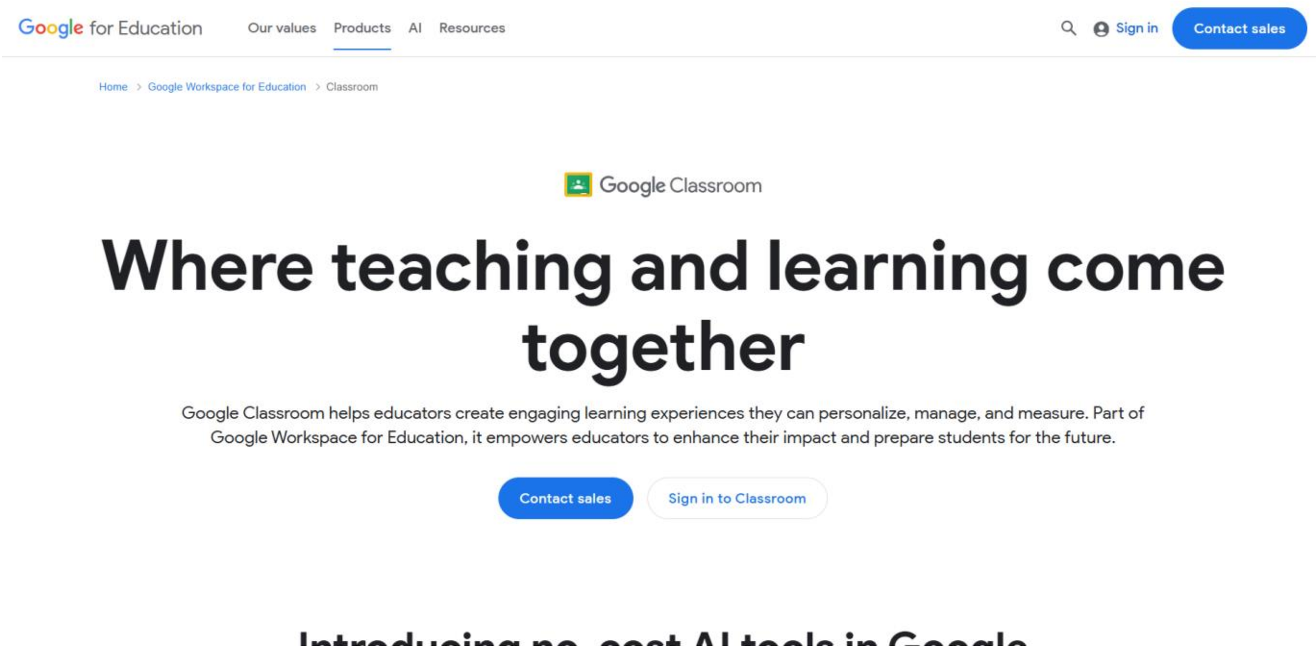
4. Moodle [4]

Introduction: Open-source LMS widely used in educational institutions.
Features: Course management, quizzes, assignments.
Limitations: Outdated UI and complex setup.



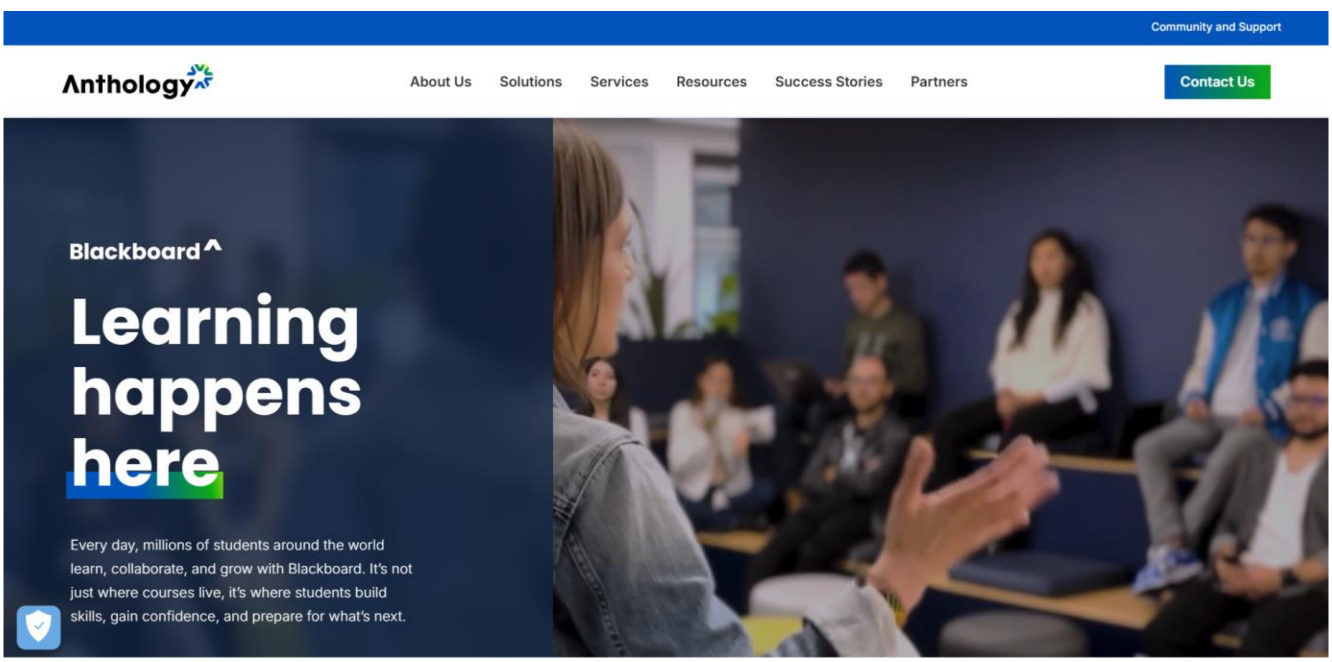
5. Google Classroom [5]

Introduction: Classroom management tool by Google.
Features: Assignment sharing, grading, announcements.
Limitations: Limited advanced learning and analytics features.



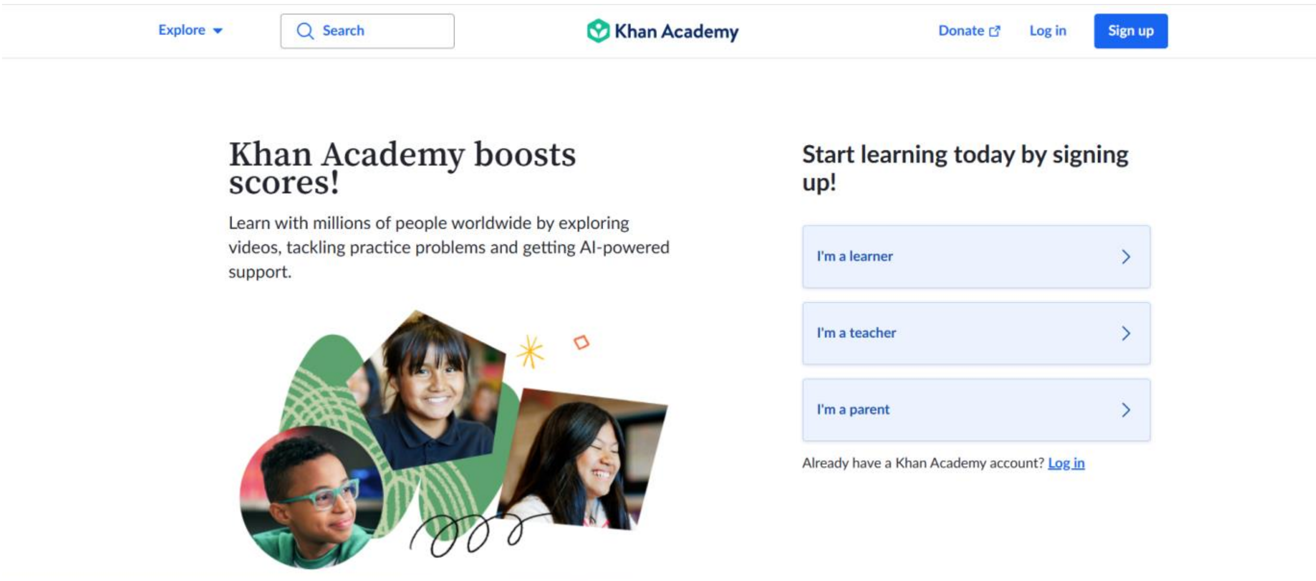
6. Blackboard [6]

Introduction: Enterprise-level LMS for universities.
Features: Course delivery, assessments, analytics.
Limitations: High cost and steep learning curve.



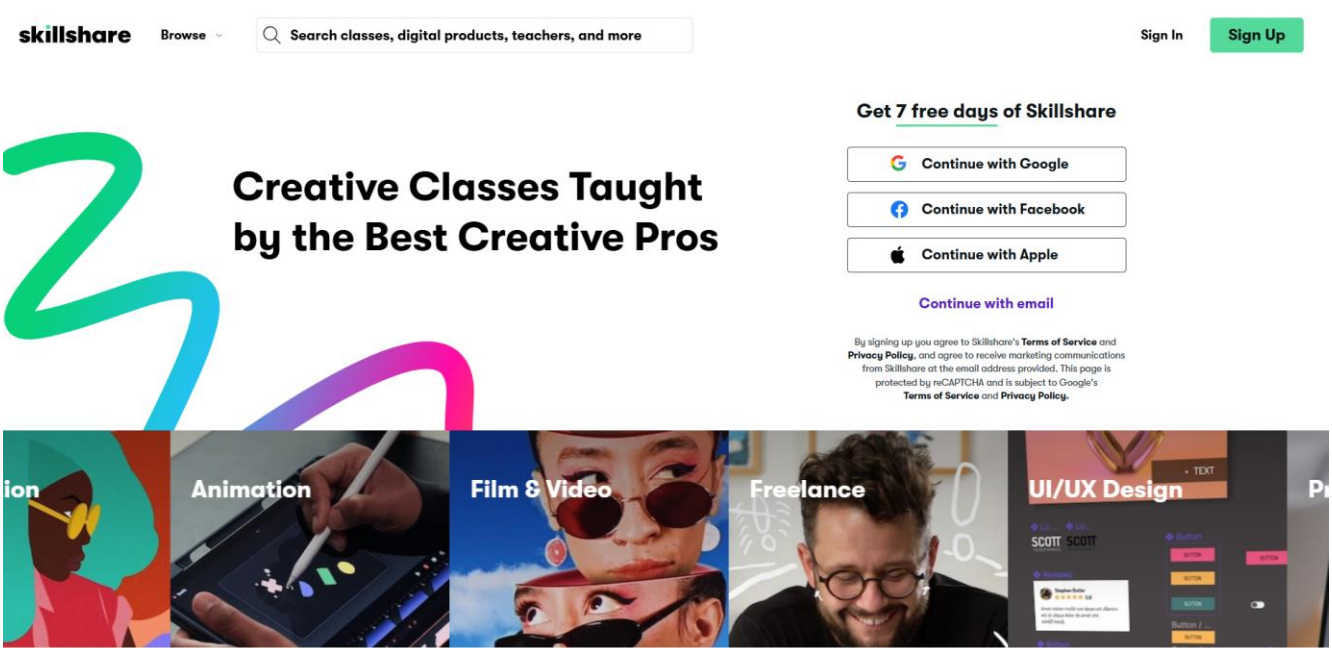
7. Khan Academy [7]

Introduction: Free learning platform for school-level education.
Features: Video lessons, practice exercises.
Limitations: Limited course variety and no educator monetization.



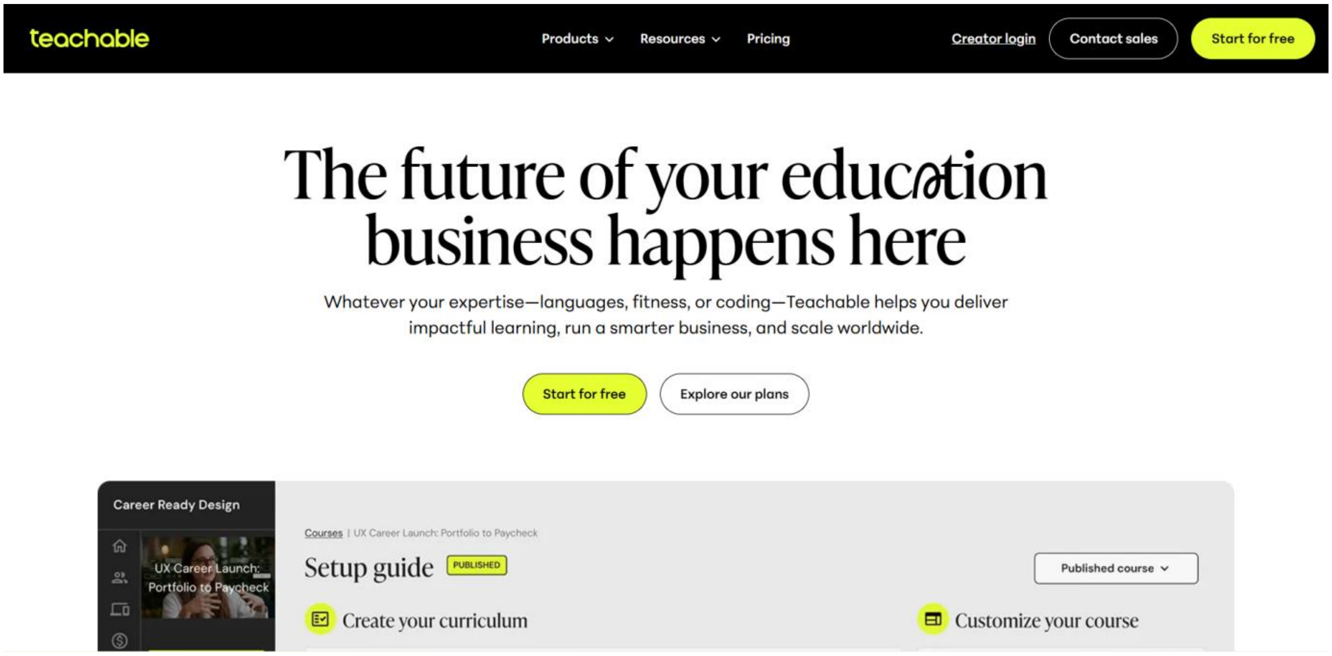
8. Skillshare [8]

Introduction: Creative learning platform for professionals.
Features: Short video classes, community learning.
Limitations: No structured assessments.



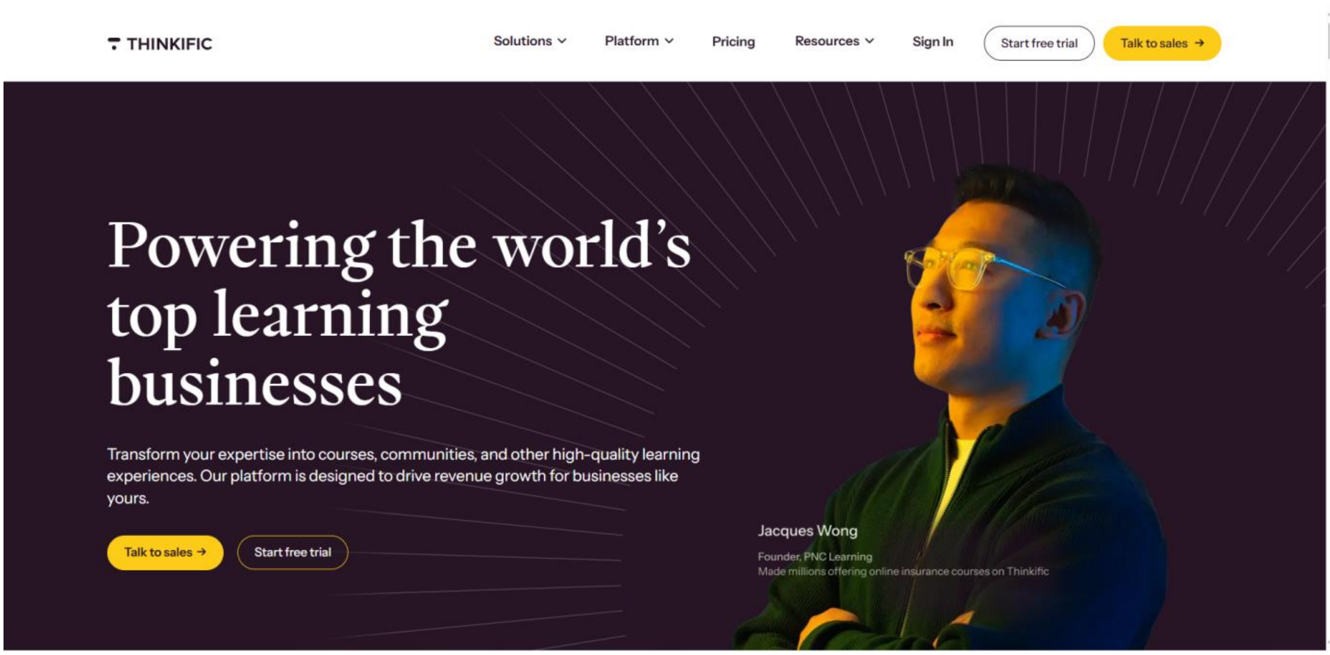
9. Teachable [9]

Introduction: Platform for instructors to create and sell courses.
Features: Course hosting, payments, analytics.
Limitations: Limited learner interaction features.



10. Thinkific [10]

Introduction: Online course creation platform.
Features: Course builder, quizzes, certificates.
Limitations: Expensive premium plans and personalization.



Limitation of Existing Literature Review:

Most existing LMS platforms lack structured learning paths, strong learner engagement features and gamification mechanisms that motivate students. Many systems also have limited mobile and offline accessibility and do not provide detailed analytics for educators. In addition, several platforms are complex or expensive, making them unsuitable for beginners and small-scale educators.

Outcomes of The Project:

1. Provides a structured and beginner friendly learning environment that guides learners step by step.
2. Improves learner engagement through gamification such as ratings, points, and badges.
3. Offers mobile responsive and easily accessible learning.
4. Supports effective course creation and management for educators through dedicated dashboards.
5. Includes testing and assessment features to evaluate learner understanding and improve learning outcomes.
6. Reduces system complexity and cost making the platform suitable for.

Tools and Technology used in The Project:

List all tools like IDEs, Technologies like framwork, you have used though out this project.

Environmental Setup:

Show the process that make sure anyone to run your project.

Project Description:

Key Features:

1. User Login and Registration: Secure authentication system for students and educators.
2. Easy Navigation Interface: Includes a navigation menu, search box, and course cards for smooth user experience.
3. Detailed Course Information: Displays course title, description, instructor name, and price.

4. Course Listing: Shows all available courses in one place for easy browsing.
5. Course Search: Allows users to search courses using keywords.
6. All Enrollments: Lists all courses that a student has enrolled in.
7. Interactive Course Player: Students can watch enrolled course videos, rate lessons, and mark them as completed, videos may play on one side of the screen.
8. Educator Dashboard: Enables instructors to view enrollments, manage courses, and monitor student activity.
9. Add Course Page: Allows educators to create and publish courses by adding lessons and content.
10. Testing and Assessment: Includes quizzes and tests to evaluate student learning.
11. Gamification: Uses points, badges, and rewards to motivate learners and increase engagement.

Course Outline:

Users register and log in to the system. Students can browse and enroll in courses, watch videos, complete lessons, and take assessments. Educators can create courses, upload content, manage students, and monitor performance through dashboards.

Discussion and Future Work:

This Learning Management System provides an easy to use and engaging platform for students and teachers. Features like course management, quizzes, and gamification make learning more interesting. In the future, AI can be used to suggest courses based on a student's interests. Advanced analytics can help teachers understand how students are doing. Real time chat can allow better communication between students and teachers.

Conclusion:

Learnify LMS provides a simple, structured, and engaging web based learning platform that addresses the limitations of existing systems. By combining accessibility, assessment and educator tools, the project enhances the online learning experience for both students and instructors.

Links

Give your live server link, github repository link and other required links.

References:

[1] Udemy, "Online Courses - Learn Anything, On Your Schedule | Udemy," *Udemy*, 2015. Available: https://www.udemy.com

[2]Coursera, "Coursera | Online Courses & Credentials by Top Educators. Join for Free," *Coursera*, 2018. https://www.coursera.org

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[4]Moodle, “Moodle - Open-source learning platform | Moodle.org,” *moodle.org*. https://moodle.org

[5] Google Classroom, “Google classroom,” *Google.com*, 2019. https://classroom.google.com

[6]“North America | Blackboard.com,” *www.blackboard.com*. https://www.blackboard.com

[7]Khan Academy, “Khan Academy,” *Khan Academy*, 2024. https://www.khanacademy.org

[8]“Online Classes by Skillshare | Start for Free Today,” *Skillshare*. https://www.skillshare.com

[9]“Teachable — Build & Sell Courses, Coaching, Memberships & More,” *Teachable.com*, 2017. https://www.teachable.com (accessed Feb. 05, 2026).

[10]“Create, Market & Sell with the #1 Online Course Platform,” *Thinkific*. https://www.thinkific.com